Unit 9 Guided Notes — Volume of 3D Solids

Cylinders · Cones · Spheres. $V = \pi r^2 h$, $V = \frac{1}{3}\pi r^2 h$, $V = \frac{4}{3}\pi r^3$.

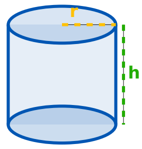
Standard: NC.8.G.9

🗐 **Blank Worksheet (PDF)** ✅ **Answer Key (PDF)** ♿ **EC/Modified Notes (PDF)**

All volume formulas use π (pi $\approx$ 3.14). Square the radius (r^2) for cylinders and cones — CUBE it (r^3) for spheres. Always include CUBIC units (cm^3, in^3, ft^3).

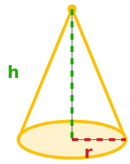
1 Key Formulas

🥫 Cylinder



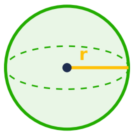
V = ______

🍦 Cone



V = ______

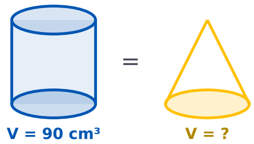
🏀 Sphere



V = ______

🔑 **THE BIG RELATIONSHIP:** A cone is exactly ______ of a cylinder with the same radius and height.

Quick Check

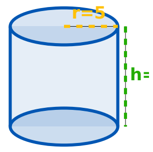


MC **The cylinder above holds 90 cm³. The cone has the SAME radius and height. What's the cone's volume?**

✓ A cone is always ⅓ of the matching cylinder: 90 ÷ 3 = 30 cm³.

2 Worked Examples

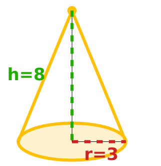
🥫 Cylinder: r = 5 cm, h = 10 cm



Substitute into $V = \pi r^2 h$: $V = \pi(5)^2(10) = \pi(25)(10)$

✓ **Answer:** $V = 250\pi \approx 785.4$ cm³

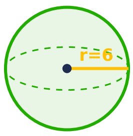
Cone: r = 3 in, h = 8 in



Substitute into $V = \frac{1}{3}\pi r^2 h$: $V = \frac{1}{3}\cdot\pi(3)^2(8) = \frac{1}{3}\cdot 72\pi$

✓ **Answer:** $V = 24\pi \approx 75.4$ in³

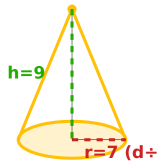
Sphere: r = 6 ft



Substitute into $V = \frac{4}{3}\pi r^3$: $V = \frac{4}{3}\cdot\pi(6)^3 = \frac{4}{3}\cdot 216\pi$

✓ **Answer:** $V = 288\pi \approx 904.3$ ft³

Cone with Diameter Given: diameter = 14 cm, h = 9 cm



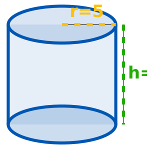
Diameter 14 → radius: $r = 7$

Substitute: $V = \frac{1}{3}\pi(7^2)(9) = \frac{1}{3}(441\pi)$

✓ **Answer:** $V = 147\pi \approx 461.8$ cm³

3 Error Analysis

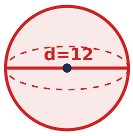
SPOT THE ERROR **Cylinder: r = 5, h = 10.**



A student wrote $V = \pi(5)(10) = 50\pi \approx 157.1$.

Forgot to SQUARE the radius! $V = \pi(5^2)(10) = 250\pi \approx 785.4$. The formula is $\pi r^2 h$.

SPOT THE ERROR **Sphere: diameter = 12.**



A student wrote $r = 12,\ V = \frac{4}{3}\pi(12^3) \approx 7238.2$.

Used DIAMETER instead of radius! $r = 6$, not 12. $V = \frac{4}{3}\pi(6^3) = 288\pi \approx 904.8$. Diameter ÷ 2 = radius.

✏ Go to Unit 9 Practice →

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